

CHE 435/525 *Process Systems Analysis and Control*
Lecture 1: **Introduction to Process Control**

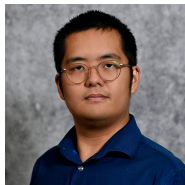
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Introduction of myself



- B.S. Chemical Engineering, Tsinghua University (2015)
 - ▶ B.S. Mathematics, Tsinghua University (2015)
- Ph.D., Chemical Engineering, Univ. of Minnesota (2020)
- Process Control Engineer, Shell Global Solutions (2020–2022)
- Assistant Professor, Chemical Engineering, NC State (2022–)
 - ▶ Affiliate Faculty, Operations Research, NC State (2024–)

A “Person in Control” for about a decade

- Past: R&D Engineer in industry, focusing on **practice**

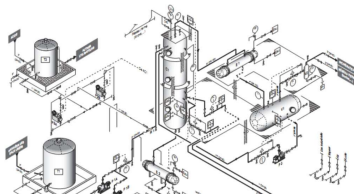
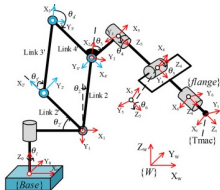


- Now: Academic investigator, focusing on **theory**
 - ▶ “Data-driven control theory” ($\approx$ control + machine learning)
 - ▶ Supported by grants from NSF, ACS, and UNC System
 - ▶ 1 postdoc + 4 Ph.D. students
- Active in the process control research community
 - ▶ American Control Conference – Associate Editor, 2025
 - ▶ AIChE Annual Meeting – Program Coordinator, 2026
 - ▶ Journal *Digital Chemical Engineering* – Associate Editor

Control: An interdisciplinary field of study

• Who studies control?

- ▶ Electrical engineers – generators, rotors, power grids
 - ★ They created this field of study first and still leads the advances
- ▶ Chemical engineers – “process control”
- ▶ Mechanical engineers – robotics, autonomous driving, etc.
- ▶ Biomedical engineers & biologists – metabolism, gene regulation
- ▶ Mathematicians & computer scientists – “control theory”
 - ★ ... can be applied to physics, chemistry, epidemics, social networks, stock markets, etc.



• Control is indispensable in all industries!

Illustrative example: Blending process

- Notation: x – mass fraction, w – mass flow rate

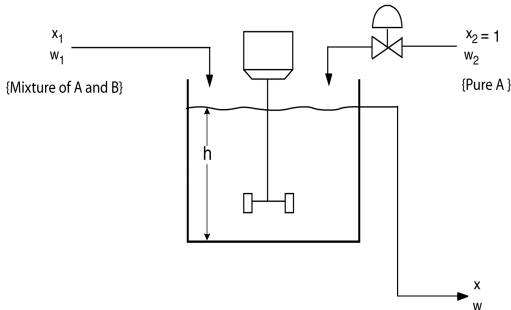


Figure 1.3. Stirred-tank blending system.

- Assumptions

- 1 $w_1 = \text{const.}$, but x_1 may fluctuate (a salty water stream to be diluted, whose saltiness is uncertain)
- 2 $x_2 = \text{const.} = 1$ (a water stream to dilute the salty water)
- 3 Perfect mixing in the tank

Control objective & Controlled variable

- Control objective is specified on some **controlled variables (CV)**.
- Typical control objectives:
 - ▶ **“Regulation”**: Keep CVs near their setpoints or within some range.
 - ▶ **“Tracking”**: Make CVs follow some desirable trajectories.
- This example: Regulate product composition (x).
- Discuss: What are the control objectives when you drive a car?



Disturbance variables & Manipulated variables

- **Disturbance Variables (DVs)**: affect CVs and cause them to deviate
 - ▶ Here: x_1 (composition of the stream to be treated)
- **Manipulated Variables (MVs)**: can be adjusted to shape the CVs
 - ▶ Here: w_2 (flow rate of the other stream)
- This Example: Keep x (CV) at a setpoint x_{sp} , despite variations in $x_1(t)$ (DV), by adjusting flow rate w_2 (MV).
- **So, how?**

From process design to process control

- Process design: Assuming all process variables are constants (i.e., x_1 is at steady state, no disturbance)

- ▶ **Steady-state relations** for design:

$$0 = \bar{w}_1 + \bar{w}_2 - \bar{w}$$

$$0 = \bar{w}_1 \bar{x}_1 + \bar{w}_2 \bar{x}_2 - \bar{w} \bar{x}$$

- ▶ To achieve desired composition (“setpoint”) $\bar{x} = x_{sp}$, we need to design w_2 as

$$\bar{w}_2 = \bar{w}_1 \frac{x_{sp} - \bar{x}_1}{1 - x_{sp}}.$$

- Process control: Adjust $w_2(t)$ against the disturbance in $x_1(t)$ (no longer const.) in order to regulate $x(t)$.

Strategy 1: Feedback control

- **Feedback control:** Measure CV and take corrective action in MV.

► For example, $w_2(t) = \bar{w}_2 + K_c[x_{sp}(t) - x(t)]$

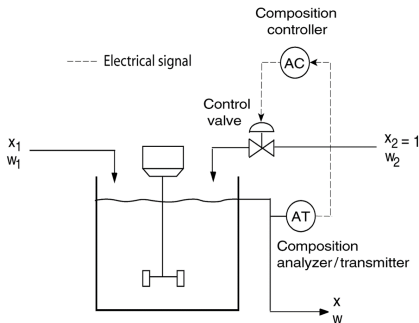


Figure 1.4. Blending system and Control Method 1.

- Very natural idea ... Think about driving a car

Strategy 2: Feedforward Control

- **Feedback control:** Measure DV and take preventive action in MV.

► For example, $w_2(t) = \bar{w}_1(t) \frac{x_{sp} - x_1(t)}{1 - x_{sp}}$

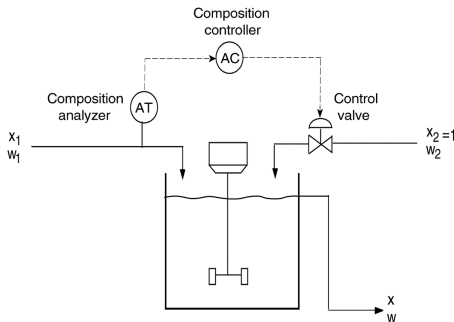


Figure 1.5. Blending system and Control Method 2.

- Discuss: Isn't this actually a better idea?

Feedback v.s. Feedforward

- Feedforward control is **idealistic**, which requires
 - ▶ accurate measurement of DV;
 - ▶ accurate model (knowledge of how DV and MV affect the CV).
- Feedback control is **realistic**.
 - ▶ Its performance/quality, however, can be restrictive
 - ▶ Goal of this course: **Optimal feedback controller design**
- If possible, combined FB/FF control will be preferred.
- Principle: To design a controller, we need the **process model**!

Roadmap

- ① Dynamic modeling of processes
 - ▶ How to describe a process as ODEs?
- ② Analysis of dynamical systems
 - ▶ How does the output respond to the input?
 - ▶ How to calculate analytically and simulate numerically?
- ③ Simple controller design & synthesis
 - ▶ How to achieve stability, robustness, and performance?
 - ▶ How to obtain the controller systematically?
- ④ Multivariable systems
 - ▶ How to deal with multi-input-multi-output?
 - ▶ How to operate the plant optimally?

“Personality” of process control

- One has to know some **theory** to do process control correctly!
 - ▶ I mean ... some **mathematics** is necessary

“A fundamental characteristic of the field of systems and control research is its **rich intersection of engineering and mathematics** – while the problem formulations stem from engineering systems, the tools and techniques employed are grounded in mathematics and computer science... It is this **abstract** feature that enables the **broad applicability** of the field.”
[F. Lamnabhi-Lagarrigue et al. (2017). *Annual Reviews in Control*, 43, 1–64.]

“An engineer who does not know mathematics never needs it. But if *he* knows it, *he* uses it frequently.” [J. Plemelj (1873–1967), mathematician]

How to get along with process control?

① Meet the basic expectations.

- ▶ Attend the classes, do homeworks, and pass exams.
- ▶ You are pursuing a degree. Minimize your distractions.

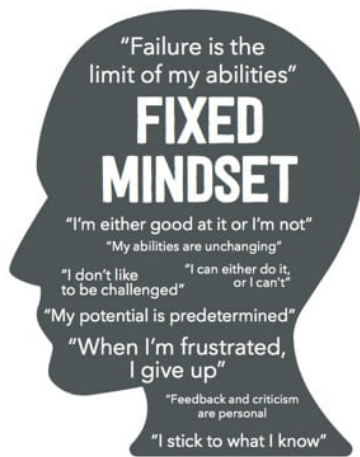
② Practice, practice, practice!

- ▶ Consent with what I write $\neq$ Ability to do it yourself.
- ▶ Fill in the “gaps” that I left on the slides (if you feel there exist).
- ▶ True understanding comes from your practice.

③ Accept the challenges.

- ▶ You will likely encounter comprehensive/complex problems in your future jobs or advanced studies.
- ▶ Maybe – you don't know the answer, you have never learned it, or your math/coding skills are not yet sufficient.
- ▶ Blame whomever that makes your life so hard, or ask for help and help yourself **proactively**
- ★ ... probably why process control takes place in your final semester

Growth mindset: What makes a good engineer



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